

$\mathbf{F}$ logically implies $\mathbf{X} \rightarrow \mathbf{Y}$ is written as $\mathbf{F} \models \mathbf{X} \rightarrow \mathbf{Y}$.

Example 6.4

$\mathbf{R} = (A, B, C, D)$ and $\mathbf{F} = \{A \rightarrow B, A \rightarrow C, BC \rightarrow D\}$, then $\mathbf{F} \models A \rightarrow D$. ■

Inference Axioms

- **F1: Reflexivity:** $(X \rightarrow X \text{ and } \mathbf{V} \subseteq Z)$
- **F2: Augmentation:** $(X \rightarrow Y) \models (XZ \rightarrow Y, \text{ and } XZ \rightarrow YZ)$
- **F3: Transitivity:** $(X \rightarrow Y \text{ and } Y \rightarrow Z) \models (X \rightarrow Z)$
- **F4: Additivity:** $(X \rightarrow Y \text{ and } X \rightarrow Z) \models (X \rightarrow YZ)$
- **F5: Projectivity:** $(X \rightarrow YZ) \models (X \rightarrow Y \text{ and } X \rightarrow Z)$
- **F6: Pseudotransitivity:** $(X \rightarrow Y \text{ and } YZ \rightarrow W) \models (XZ \rightarrow W)$

Inference rules **F1** through **F3** are variations of the Armstrong axioms, so called after W. W. Armstrong, who first proposed them (Arms 74). Example 6.5 gave informal arguments showing that each of the inference axioms **F1** through **F6** is sound (i.e., correct). This means that whenever an FD $X \rightarrow Y$ can be derived from a set of FDs **F** using these axioms, then $F \models X \rightarrow Y$. It has been shown that the converse also holds, even for the subset **F1** through **F3**. Whenever $F \models X \rightarrow Y$, $X \rightarrow Y$ can be derived from **F** using these inference axioms. These axioms form a complete axiom system for FDs. Rules **F4** through **F6** in particular can be derived from rules **F1** through **F3**.

Figure BRelation R on the scheme $R(A, B, C, D, E)$.

R	A	B	C	D	E
	a ₁	b ₁	c ₂	d ₁	e ₁
	a ₂	b ₂	c ₁	d ₂	e ₂
	a ₃	b ₁	c ₂	d ₁	e ₃
	a ₃	b ₃	c ₃	d ₃	e ₄
	a ₁	b ₂	c ₁	d ₂	e ₅
	a ₄	b ₄	c ₄	d ₄	e ₆
	a ₃	b ₂	c ₁	d ₂	e ₇
	a ₅	b ₄	c ₄	d ₄	e ₈

In Figure B, the FD $B \rightarrow C$ is satisfied and by augmentation we find that the FDs $AB \rightarrow C$, $BC \rightarrow C$, $BD \rightarrow C$, $BE \rightarrow C$ and $ABC \rightarrow C$, $BCD \rightarrow C$, etc. are also satisfied.

We note from Figure B that the FDs $B \rightarrow C$, $B \rightarrow D$, and consequently $B \rightarrow CD$ are all satisfied.

Projectivity: This axiom is the inverse of the additivity axiom; it splits up or projects an FD whose right-hand side is a union of attributes into a number of FDs. Each projected FD has the same left-hand side as the original FD and each contains a subset of the original right-hand side.

For the relation R of Figure B, the FD $B \rightarrow CD$ is satisfied and hence, by projectivity, $B \rightarrow C$ and $B \rightarrow D$.

Transitivity: For the relation R of Figure B, the FDs $B \rightarrow C$ and $C \rightarrow D$ are satisfied and hence, by transitivity, $B \rightarrow D$. Thus, when the value for B is b_1 in R, the value of C is c_2 . Similarly, when the value of C is c_2 , the value of D is d_1 . When the value of B is b_1 , the value of D is d_1 .

Pseudotransitivity: This axiom follows from axioms **F2** and **F3**. Given $X \rightarrow Y$, by **F2**, $XZ \rightarrow YZ$ and since $YZ \rightarrow W$ is given, then by **F3**, $XZ \rightarrow W$.

The relation R of Figure B satisfies the FDs $C \rightarrow B$ and $AB \rightarrow E$, so by pseudotransitivity, the FD $CA \rightarrow E$ is also satisfied. ■

6.4.2 Closure of a Set of Functional Dependencies

The set of functional dependencies that is logically implied by F is called the **closure** of F and is written as F^+ .

Definition: If F is a set of FDs on a relation scheme R , then F^+ , the closure of F , is the smallest set of FDs such that² $F^+ \supseteq F$ and no FD can be derived from F by using the inference axioms that are not contained in F^+ . If R is not specified, it is assumed to contain all the attributes that appear in F .

F^+ is the set of FDs that are implied by the FDs in F , i.e., $F^+ = \{X \rightarrow Y \mid F \models X \rightarrow Y\}$.

An FD f in F^+ is logically implied by F since any relation R on the relation scheme R that satisfies the FDs in F also satisfies the FD in F^+ and, hence, f .

² $F^+ \supseteq F$ denotes that F^+ contains F .

FD Closure

Example 6.6

Let $R = (A, B, C, D)$ and $F = \{A \rightarrow B, A \rightarrow C, BC \rightarrow D\}$. Since $A \rightarrow B$ and $A \rightarrow C$, then by **F4** $A \rightarrow BC$. Now since $BC \rightarrow D$, then by **F3** $A \rightarrow D$, i.e., $F \models A \rightarrow D$ and thus $A \rightarrow D$ is in F^+ . ■

An example of an FD not implied by a given set of FDs is illustrated below.

Example 6.7

Let $F = \{W \rightarrow X, X \rightarrow Y, W \rightarrow XY\}$. Then F^+ includes the set $\{W \rightarrow W, X \rightarrow X, Y \rightarrow Y, W \rightarrow X, X \rightarrow Y, W \rightarrow XY, W \rightarrow Y\}$. The first three FDs follow from axiom **F1**; the next three FDs are in F and hence in F^+ . Since $W \rightarrow XY$, then by axiom **F5**, $W \rightarrow X$ and $W \rightarrow Y$. However, F^+ does not contain an FD, e.g., $W \rightarrow Z$, because Z is not contained in the set of attributes that appear in F . ■

Algorithm

6.1

Algorithm to Compute X^+

Input: A set of functional dependencies F and a set of attributes X .

Output: The closure X^+ of X under the FDs in F .

```
 $X^+ := X$ ; (* initialize  $X^+$  to  $X$  *)
change := true;
while change do
  begin
    change := false;
    for each FD  $W \rightarrow Z$  in  $F$  do
      begin
        if  $W \subseteq X^+$  then do
          begin
             $X^+ := X^+ \cup Z$ ;
            change := true;
          end
        end
      end
    end
  end
  (*  $X^+$  now contains the closure of  $X$  under  $F$  *)
```

Example 6.8

Let $X = BCD$ and $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC\}$. We want to compute the closure X^+ of X under F . We initialize X^+ to X , i.e., $X^+ := BCD$. Now since the left-hand side of the FD $CD \rightarrow E$ is a subset of X^+ , i.e., $CD \subseteq X^+$, X^+ is augmented by the right-hand side of the FD, i.e., E ; thus X^+ now becomes equal to $BCDE$. Similarly, since $D \subseteq X^+$, the right-hand side of the FD $D \rightarrow AEH$ is added to X^+ , which now becomes $ABCDEH$. X^+ cannot be augmented any further and Algorithm 6.1 ends with X^+ equal to $ABCDEH$. ■

Algorithm

6.2

Membership Algorithm

Input: A set of functional dependencies F and the functional dependency $X \rightarrow Y$.

Output: Is $X \rightarrow Y \in F^+$ or not?

Compute X^+ using Algorithm 6.1.

if $Y \subseteq X^+$ then $X \rightarrow Y \in F^+ := \text{true}$;

6.4.4

Testing if an FD is in a Closure

As mentioned earlier, to find out whether $F \models X \rightarrow Y$ without computing F^+ requires the computation of X^+ under the set of FDs F , and if $Y \subseteq X^+$ then F logically implies the functional dependency $X \rightarrow Y$, otherwise it does not. Algorithm 6.2 tests the membership of $X \rightarrow Y$ in F^+ by this indirect scheme. It uses Algorithm 6.1 to compute the closure of X under F .

Example 6.9

Let $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC\}$. We want to find if $F \models BCD \rightarrow H$.

Having computed BCD^+ as being $ABCDEH$ we can clearly see that the FD $BCD \rightarrow H$ is implied by the FD F since $H \subseteq BCD^+$. ■

Consider the following relation scheme pertaining to the information about a student maintained by an university:

STDINF(*Name*, *Course*, *Phone_No*, *Major*, *Prof*, *Grade*)

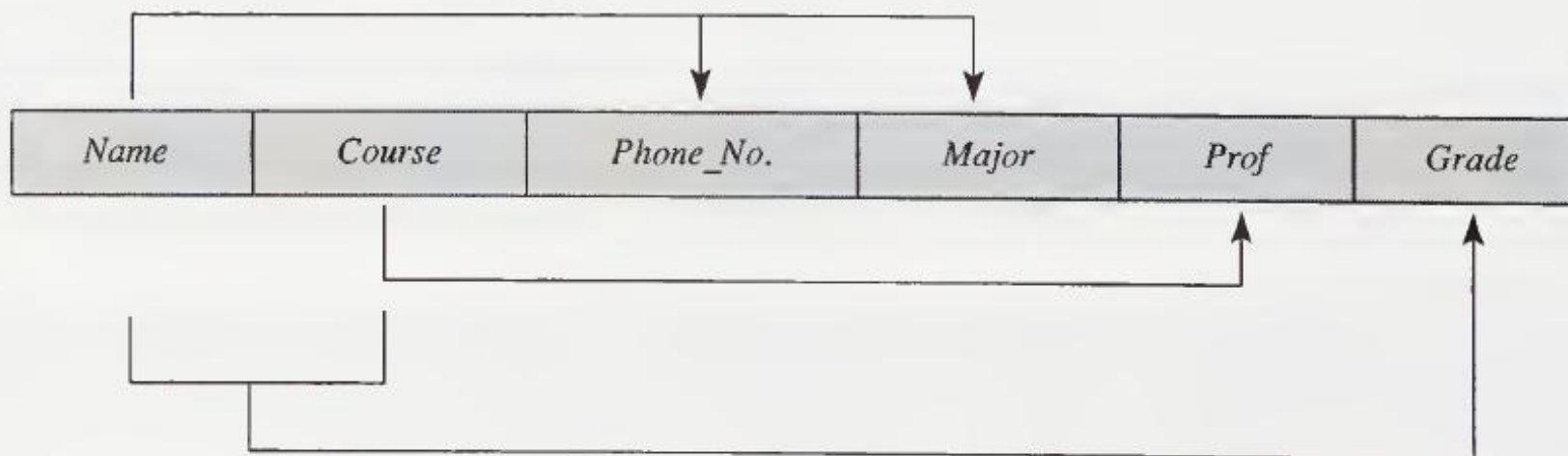
Figure 6.1 shows some tuples of a relation on the relation scheme **STDINF** (*Name*, *Course*, *Phone_No*, *Major*, *Prof*, *Grade*). The functional dependencies¹ among its attributes are shown in Figure 6.2. The key of the relation is *Name Course* and the relation has, in addition, the following functional dependencies $\{Name \rightarrow Phone_No, Name \rightarrow Major, Name\ Course \rightarrow Grade, Course \rightarrow Prof\}$.

Here the attribute *Phone_No*, which is not in any key of the relation scheme **STDINF**, is not functionally dependent on the whole key but only on part of the key, namely, the attribute *Name*. Similarly, the attributes *Major* and *Prof*, which are not in any key of the relation scheme **STDINF** either, are fully functionally dependent on the attributes *Name* and *Course*, respectively. Thus the determinants of these functional dependencies are again not the entire key but only part of the key of the relation. Only the attribute *Grade* is fully functionally dependent on the key *Name Course*.

Figure 6.1 Student data represented in relation STDINF.

<i>Name</i>	<i>Course</i>	<i>Phone_No</i>	<i>Major</i>	<i>Prof</i> ;	<i>Grade</i>
Jones	353	237-4539	Comp Sci	Smith	A
Ng	329	427-7390	Chemistry	Turner	B
Jones	328	237-4539	Comp Sci	Clark	B
Martin	456	388-5183	Physics	James	A
Dulles	293	371-6259	Decision Sci	Cook	C
Duke	491	823-7293	Mathematics	Lamb	B
Duke	356	823-7293	Mathematics	Bond	in prog
Jones	492	237-4539	Comp Sci	Cross	in prog
Baxter	379	839-0827	English	Broes	C

Figure 6.2 Function dependencies in STDINF.



The problems in the relation scheme **STDINF** can be resolved if we replace it with the following relation schemes:

STUDENT_INFO (*Name, Phone_No, Major*)

TRANSCRIPT (*Name, Course, Grade*)

TEACHER (*Course, Prof*)

Note that the decomposition of **STDINF** into the relation schemes **STUDENT**(*Name, Phone_No, Major, Grade*) and **COURSE**(*Course, Prof*) is a bad decomposition for the following reasons:

1. Redundancy and update anomaly, because the data for the attributes *Phone_No* and *Major* are repeated.
2. Loss of information, because we lose the fact that a student has a given grade in a particular course.

6.4.5 Covers

Given a set of FDs $\mathbf{F}$, $\mathbf{F}^+$ is the closure of $\mathbf{F}$ and contains all FDs that can be derived from $\mathbf{F}$. As mentioned earlier, $\mathbf{F}^+$ can be very large; hence, we will look for a smaller set of FDs that are representative of the closure of $\mathbf{F}$. Suppose we have another set of FDs $\mathbf{G}$. We say that $\mathbf{F}$ and $\mathbf{G}$ are equivalent if the closure of $\mathbf{F}$ is identically equal to the closure of $\mathbf{G}$, i.e., $\mathbf{F}^+ = \mathbf{G}^+$. If the sets of FDs $\mathbf{F}$ and $\mathbf{G}$ are equivalent, we can consider one to be representative of the other or one **covers** the other. Thus $\mathbf{F}$ covers $\mathbf{G}$ and $\mathbf{G}$ covers $\mathbf{F}$.

Definition: Given two sets of FDs $\mathbf{F}$ and $\mathbf{G}$ over a relation scheme $\mathbf{R}$, $\mathbf{F}$ and $\mathbf{G}$ are equivalent (i.e., $\mathbf{F} \equiv \mathbf{G}$) if the closure of $\mathbf{F}$ is identically equal to the closure of $\mathbf{G}$ (i.e., $\mathbf{F}^+ = \mathbf{G}^+$). If $\mathbf{F}$ and $\mathbf{G}$ are equivalent, then $\mathbf{F}$ **covers** $\mathbf{G}$ and $\mathbf{G}$ **covers** $\mathbf{F}$.

Algorithm

6.3

Nonredundant Cover

Input: A set of FDs F .

Output: A nonredundant cover of F .

```

G := F; (* initialize G to F *)
for each FD  $X \rightarrow Y$  in G do
    if  $X \rightarrow Y \in \{F - (X \rightarrow Y)\}^+$  (* i.e.,  $\{F - (X \rightarrow Y)\} \models X \rightarrow Y$  *)
    then F :=  $\{F - (X \rightarrow Y)\}$ ;
G := F; (* G is the nonredundant cover of F *)
end;
```

Example 6.10

If $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC\}$ then the FDs $CD \rightarrow E$ and $DH \rightarrow BC$ are redundant. We find that $(CD)^+$ under $\{F - (CD \rightarrow E)\}$ is equal to $ABCDEH$, and since the right-hand side of the FD $(CD \rightarrow E) \subseteq (CD)^+$ under $\{F - (CD \rightarrow E)\}$, $\{F - (CD \rightarrow E)\} \models CD \rightarrow E$. We now remove this redundant FD from F and then find that for the FD $(DH \rightarrow BC)$, $(DH)^+$ under $\{F - (DH \rightarrow BC)\}$ is $ABCDEH$. Since the right-hand side of the FD $(DH \rightarrow BC) \subseteq (DH)^+$, the FD $(DH \rightarrow BC)$ is also redundant. No remaining FDs can be removed from the modified F . Thus a nonredundant cover for F is $\{A \rightarrow BC, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD\}$. ■

If F is a set of FDs and if G is a nonredundant cover of F , then it is not true that G has the minimum number of FDs. In fact, there may exist a cover G' of F that has fewer FDs than G . Thus, a minimum cover G' of F has as small a number of FDs as any other cover of F . It is needless to add that a minimum cover G' of F has no redundant FDs; however, a nonredundant cover of F need not be minimal, as we see in Example 6.11. We will not discuss an algorithm to derive a minimum cover in this text. The interested reader is referred to the bibliographic notes at the end of the chapter.

A set of functional dependencies F_c is a **canonical cover** if every FD in F_c satisfies the following:

1. Each FD in F_c is *simple*. Recall that in a simple FD the right-hand side has a single attribute, i.e., each FD is of the form $X \rightarrow A$.

2. For no FD $X \rightarrow A$ with $Z \subset X$ is $\{(F_c - (X \rightarrow A)) \cup (Z \rightarrow A)\} \models F_c$. In other words, the left-hand side of each FD does not have any extraneous attributes, or the FDs in F_c are left reduced.
3. No FD $X \rightarrow A$ is redundant, i.e., $\{F_c - (X \rightarrow A)\}$ does not logically imply F_c .

A canonical cover is sometimes called **minimal**.

Given a set F of functional dependencies we can find a canonical set F_c ; obviously F_c covers F .

Example 6.11

If $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC\}$, then a nonredundant cover for F is $\{A \rightarrow BC, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD\}$. The FD $ABH \rightarrow BD$ can be decomposed into the FDs $ABH \rightarrow B$ and $ABH \rightarrow D$. Now, since the FD $A \rightarrow B$ is in F , we can left reduce these decomposed FDs into $AH \rightarrow B$ and $AH \rightarrow D$. We also notice that $AH \rightarrow B$ is redundant since the FD $A \rightarrow B$ is already in F . This gives us the canonical cover as $\{A \rightarrow B, A \rightarrow C, E \rightarrow C, D \rightarrow A, D \rightarrow E, D \rightarrow H, AH \rightarrow D\}$. ■

Note that if F_c is a canonical cover and if we form G using the additivity axiom (such that the FDs with the same left-hand sides are merged into a single FD with the right-hand sides combined), then F_c and G are equivalent. However, G will contain nonsimple FDs.

6.4.8 Functional Dependencies and Keys

Earlier we discussed the concept of uniquely identifying an entity within an entity set by a key, the key being a set of attributes of the entity. A relation scheme $\mathbf{R}$ has a similar concept, which can be explained using functional dependencies.

Definition: Given a relation scheme $\mathbf{R} \{A_1 A_2 A_3 \dots A_n\}$ and a set of functional dependencies $\mathbf{F}$, a key of $\mathbf{R}$ is a subset of $\mathbf{R}$ such that $\mathbf{K} \rightarrow A_1 A_2 A_3 \dots A_n$ is in $\mathbf{F}^+$ and for any $\mathbf{Y} \subset \mathbf{K}$, $\mathbf{Y} \rightarrow A_1 A_2 A_3 \dots A_n$ is not in $\mathbf{F}^+$.

The first requirement indicates that the dependency of all attributes of $\mathbf{R}$ on $\mathbf{K}$ is given explicitly in $\mathbf{F}$ or can be logically implied from $\mathbf{F}$. The second requirement indicates that no proper subset of $\mathbf{K}$ can determine all the attributes of $\mathbf{R}$. Thus, the key used here is minimal with respect to this property and the FD $\mathbf{K} \rightarrow \mathbf{R}$ is left reduced. A superset of $\mathbf{K}$ can then be called a superkey.

If there are two or more subsets of $\mathbf{R}$ such that the above conditions are satisfied, such subsets are called candidate keys. In such a case one of the candidate keys is designated as the primary key or simply as the key.

We do not allow any attribute in the key of a relation to have a null value.

Example 6.12

If $\mathbf{R} (ABCDEH)$ and $\mathbf{F} = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC\}$, then CD is a key of $\mathbf{R}$ because $CD \rightarrow ABCDEH$ is in $\mathbf{F}^+$ (since $(CD)^+$ under $\mathbf{F}$ is equal to $ABCDEH$ and $ABCDEH \subseteq ABCDEH$). Other candidate keys of $\mathbf{R}$ are AD and ED . ■

Full Functional Dependency

The concept of left-reduced FDs and fully functionally dependency is defined below and illustrated in Example 6.13.

Definition: Given a relational scheme R and an FD $X \rightarrow Y$, Y is **fully functionally dependent** on X if there is no Z , where Z is a proper subset of X such that $Z \rightarrow Y$. The dependency $X \rightarrow Y$ is left reduced, there being no extraneous attributes in the left-hand side of the dependency.

Example 6.13

In the relation scheme $R(ABCDEH)$ with the FDs $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow C, CD \rightarrow AH, ABH \rightarrow BD, DH \rightarrow BC\}$, the dependency $A \rightarrow BC$ is left reduced and BC is fully functionally dependent on A . However, the functional dependency $ABH \rightarrow D$ is not left reduced, the attribute B being extraneous in this dependency. ■

Prime Attribute and Nonprime Attribute

We defined the key of a relation scheme earlier. We distinguish the attributes that participate in any such key as indicated by the following definition.

Definition: An attribute A in a relation scheme R is a **prime attribute** or simply **prime** if A is part of any candidate key of the relation. If A is not a part of any candidate key of R , A is called a **nonprime attribute** or simply **nonprime**.

Example 6.14

If $R(ABCDEH)$ and $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow C, AH \rightarrow D\}$, then AH is the only candidate key of R . The attributes A and H are prime and the attributes B, C, D , and E are nonprime. ■

